

PRST (15/03/2021)

Strong law of large number

If $\{X_n\}$ is a sequence of r.v. independent and identically distributed (i.i.d.) with $E(X_n) = \mu \forall n$
 $\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mu$ (a.s. - almost surely)

Central limit theorem:

Let $\{X_n\}$ be a seq. of r.v.s independent and identically (i.i.d.) distributed $E(X_n) = \mu \forall n$
 $\text{Var}(X_n) = \sigma^2 \forall n, (\sigma > 0)$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n$$

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \leq x_0\right) = \int_{-\infty}^{x_0} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$P(X \leq x_0) = \int_{-\infty}^{x_0} f(t) dt$$

$$P(Z \leq x_0) = \int_{-\infty}^{x_0} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$Z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

Ex. A computer has a random number generator that generates number between 0 and 1.
 Suppose we run it 100 times

$$P(S_{100} \geq 52) = ?$$

(S_{100} : sum of 100 numbers)

X_n : denoting the number generated by the computer at n th click.

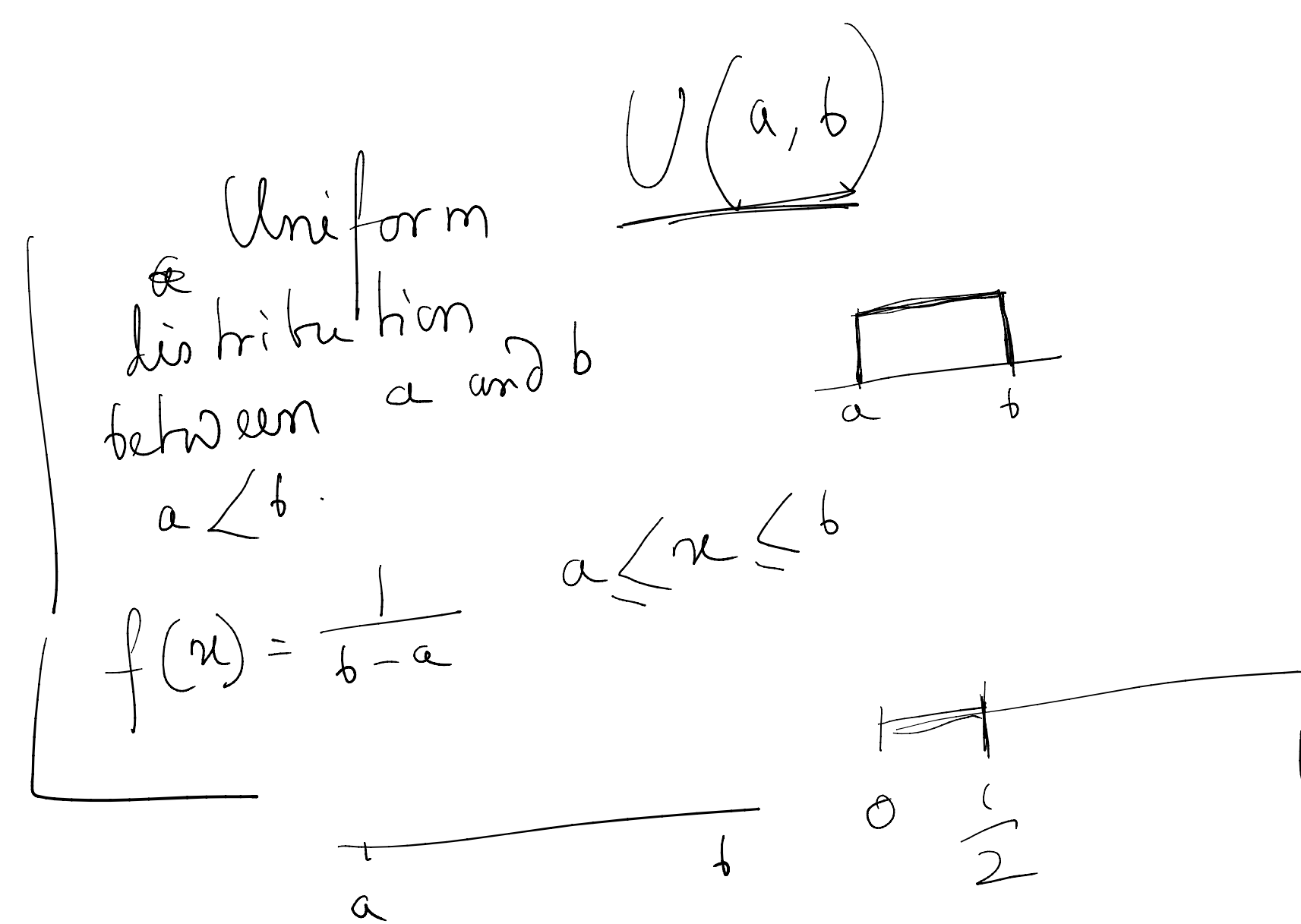
$$S_{100} = \sum_{i=1}^{100} X_i$$

$$X_n \sim U(0,1) \forall n$$

$$\mu = E(X_n) = \frac{1}{2}$$

$$\sigma^2 = \text{Var}(X_n) = \frac{1}{12}$$

$$\sigma = \sqrt{\text{Var}(X_n)}$$



$$f(x) = 1 \quad 0 \leq x \leq 1$$

Using CLT

$$Z = \frac{\frac{S_{100}}{100} - E(X_n)}{\sqrt{\frac{\text{Var}(X_n)}{n}}} \sim N(0,1)$$

$$P(S_{100} \geq 52)$$

$$= P\left(\frac{\frac{S_{100}}{100} - \frac{1}{2}}{\sqrt{\frac{1}{12 \times 100}}} \geq \frac{\frac{52}{100} - \frac{1}{2}}{\sqrt{\frac{1}{12 \times 100}}}\right)$$

$$= P(Z \geq c) = \int_c^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$= 1 - \Phi(c)$$

$$X \sim N(0,1)$$

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, -\infty < x < \infty$$

$$P(X \leq x_0) = \Phi(x_0) = \int_{-\infty}^{x_0} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$c = \frac{\frac{52}{100} - \frac{1}{2}}{\sqrt{\frac{1}{12 \times 100}}}$$

Eg By clinical study it is found that people over age 50 had a mean cholesterol 54, sd 17.

Suppose a physician has 40 patients over age 50 and wants to determine the prob that the mean cholesterol of 40 patients is 60 ~~and~~ or more.

Sol X_i — cholesterol of i th patient
 $E(X_i) = 54$, $\text{Var}(X_i) = 17^2$

$$P\left(\frac{1}{40} \sum_{i=1}^{40} X_i \geq 60\right)$$